Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Course: B.Tech | *Time: 03 Hours*
Code: 104505 | *Subject: Microprocessors and Microcontrollers* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which interrupt has the highest priority in 8085?
(i) RST 5.5 (iii) RST 6.5
(ii) INTR (iv) TRAP

(b) Which signal indicates completion of data transfer?
(i) READY (iii) ALE
(ii) IO/M (iv) SOD

(c) The 8255 has how many ports?
(i) 2 (iii) 3
(ii) 4 (iv) 5

(d) Which mode of 8255 is used for simple I/O operations?
(i) Mode 0 (iii) Mode 1
(ii) Mode 2 (iv) BSR mode

(e) Sample-and-Hold circuit is mainly used with
(i) DAC (iii) ADC
(ii) LCD (iv) Stepper Motor

(f) Which controller is used for keyboard interfacing?
(i) 8255 (iii) 8279
(ii) 8253 (iv) 8251

(g) The 8086 has how many general-purpose registers?
(i) 2 (iii) 4
(ii) 8 (iv) 16

(h) Which register stores the address of next instruction?
(i) AX (iii) BX
(ii) IP (iv) DX

(i) Which processor has on-chip cache memory?
(i) 8086 (iii) 80286
(ii) 80386 (iv) 80486

(j) Which instruction changes program sequence?
(i) MOV (iii) ADD
(ii) JMP (iv) ANL

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Draw and explain the architecture of the 8085 microprocessor.

(b) Discuss the timing diagram and machine cycle of the 8085.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Explain different modes of operation of the 8255 PPI.

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(b) Explain the architecture and operation of the 8251 USART.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the architecture and operation of Intel 8255.

(b) Explain memory organization in the 8086 microprocessor.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Draw and explain the architecture of the 8086 microprocessor.

(b) Discuss the pin diagram and functions of the 8086 microprocessor.

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain different addressing modes of the 8086 with examples.

(b) Explain the data transfer instructions of the 8086 with examples.

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) Discuss stack operations and subroutine calls in the 8086.

(b) Write an assembly language program for addition of two 16-bit numbers.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Discuss memory organization in the 8051 microcontroller.

(b) Discuss logic instructions and control transfer instructions in the 8051.

Q.9 **Write short note on any four of the following:** **[3½ x 4 = 14]**

(a) Timing diagram

(b) 8251 USART

(c) 8086 Addressing modes

(d) PUSH and POP instructions

(e) Differentiate between hardware interrupts and software interrupts

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